

Ibrahim Abiodun Imam

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SUMMARY

PhD candidate in Chemical Engineering specializing in computational chemistry and AI-enabled molecular design. Experienced in structure-based modeling (docking, MD simulation, enhanced sampling), cheminformatics (RDKit and OpenEye), and high-throughput workflows on HPC. Industry co-op experience supporting DNA-encoded library (DEL) hit triage and small-molecule design in collaboration with medicinal chemistry teams.

EDUCATION

PhD, Chemical Engineering | University of Kentucky | Aug 2022 - Nov 2026 (Expected)

GPA: 3.92

TECHNICAL SKILLS

- Cheminformatics: OpenEye (ROCS, OMEGA), RDKit, Open Babel, Spotfire.
- Molecular modeling: GROMACS, OpenMM, AutoDock Vina, ORCA, PyMOL
- Enhanced sampling: PLUMED (well-tempered metadynamics, umbrella sampling), replica-exchange concepts
- Free energy & scoring: TI, MBAR, MM/GBSA
- Machine learning: Mckit-learn, ensemble learning, PyTorch; data prep and model validation
- Programming: Python (NumPy, Pandas, MDAnalysis), R, Linux/Bash, Git

EXPERIENCE

Cheminformatics Co-op (DNA-Encoded Libraries) | GlaxoSmithKline (GSK) | Cambridge MA | Jan 2026 - Present

- Developed ML ensembles to identify target-specific hits from large-scale chemical libraries.
- Supported DEL selection data analysis and hit triage using Python/RDKit workflows (enrichment, clustering, substructure analytics).
- Prepared interactive Spotfire dashboards for medicinal chemistry teams to support series prioritization and SAR-style visualizations.
- Prepared reports for medicinal chemistry teams: series prioritization, analog suggestions, and SAR-style visualizations.

Graduate Research Assistant | University of Kentucky | Lexington, KY | Aug 2022 - Present

- Developed a multi-scale pipeline combining molecular dynamics simulations with deep learning (PyTorch) to study protein structure-function relationships; resulted in 2 peer-reviewed publications.
- Curated and preprocessed small molecule datasets for docking, and ML training; improved model precision by 10%.
- Performed computational experiments to investigate biophysical mechanisms in proteins.
- Automated HPC workflows with Python and Bash, reducing simulation setup time by 45%.
- Collaborated with interdisciplinary teams to derive insights into EGFR drug discovery efforts.

Graduate Research Fellow | Berkeley National Laboratory | Berkeley, CA | Jun 2023 - Aug 2023

- Enhanced NWChemEx capabilities for biochemical system modeling and improved scalability on HPC resources.
- Built automation scripts for benchmarking and regression testing to validate software changes.
- Conducted benchmarking analyses and documented recommendations for maintainable workflows.

SELECTED PROJECTS & RESEARCH HIGHLIGHTS

- EGFR mutation resistance: MD + quantum chemical calculations to characterize resistance mechanisms against osimertinib (Publication: Proteins: Structure, Function and Bioinformatics 2025).
- Functional peptides: integrated protein language models with MD to discover antifreeze and antibiofouling peptides (Publications: J Mater Chem B 2026 and Langmuir, 2024).

CORE COMPETENCIES

- Structure-based drug design: docking, binding pose analysis, MD-driven binding hypotheses, conformational analysis
- Cheminformatics & data: RDKit pipelines, similarity search, clustering, hit triage, dataset curation
- Physics-based methods: enhanced sampling (metadynamics, umbrella sampling), free energy methods (TI, MBAR, MM/GBSA)
- AI/ML for molecules: feature engineering, QSAR, deep learning (PyTorch), model evaluation and deployment-ready scripting
- Scientific computing: Python (NumPy, Pandas, MDAnalysis), Linux/Bash, Git, HPC job automation

SELECTED PUBLICATIONS

- Imam I.A., et al. Discovering naturally occurring antifreeze peptides from microbiome by integrating protein language models and molecular dynamics simulation. *J. Mater Chem B* (2026)
- Imam I.A., et al. In-Silico Predictions of Drug Resistance in Lung Cancers with EGFR Mutation. *PASC '25: Proceedings of the Platform for Advanced Scientific Computing Conference* (2025)
- Imam, I. A., et al. L858R/L718Q and L858R/L792H mutations of EGFR inducing resistance against osimertinib. *Proteins: Structure, Function, and Bioinformatics* (2025).
- Imam, I. A., et al. Integrating protein language model and molecular dynamics simulations to discover anti-biofouling peptides. *Langmuir* (2024).

TRAINING & WORKSHOPS

- Virtual School on Machine Learning for Molecules (Apr 2025): data cleaning, molecular featurization, ML/DL
- MDAnalysis/MolSSI Workshop, Arizona State University (Jun 2024): scientific software development and version control
- NSF i-CoMSE Methods for Enhanced Sampling (Apr 2023): umbrella sampling, metadynamics, replica exchange